AgroScan - An Interactive SOC Prediction Tool for Farmers

Supplementary Material

August 13, 2026

1 Feature Transformation

The vegetation- and soil indices summarized in Tab. 1 were derived from the S-2 band values following the approach by [1].

Table 1: Predictor features and their corresponding formulas. Adapted from Nguyen et al. (2022).

Predictor feature	**Acronym**	**Formula**
Ratio Vegetation Index	RVI	NIR/Red
Normalized Difference Vegetation Index	NDVI	$\frac{\text{NIR}-\text{Red}}{\text{NIR}+\text{Red}}$
Green Normalized Difference Vegetation Index	GNDVI	$\frac{\text{NIR}-\text{Green}}{\text{NIR}+\text{Green}}$
Normalized Difference Index using Bands 4 & 5 of S-2	NDI45	$\frac{\text{RE1}-\text{Red}}{\text{RE1}+\text{Red}}$
Soil Adjusted Vegetation Index	SAVI	$\frac{(1+L)(\text{NIR}-\text{Red})}{L+\text{NIR}+\text{Red}},\ L=0.5$
Inverted Red-Edge Chlorophyll Index	IRECI	$\frac{\text{NIR}-\text{Red}}{\text{RE1}/\text{RE2}}$
Modified Chlorophyll Absorption in Reflectance Index	MCARI	$[(\text{RE1}-\text{Red})-0.2(\text{RE1}-\text{Green})]\cdot\frac{\text{RE1}}{\text{NIR}}$
Brightness Index	BI	$\sqrt{\frac{\text{Red}^2+\text{Green}^2}{2}}$
Brightness Index 2	BI2	$\sqrt{\frac{\text{Red}^2+\text{Green}^2+\text{NIR}^2}{3}}$
Redness Index	RI	$\frac{\text{Red}^2}{\text{Green}^2}$
Color Index	CI	$\frac{\text{Red}-\text{Green}}{\text{Red}+\text{Green}}$
Month (sin)	`sin_month`	$\sin\left(\frac{2\pi\cdot\text{month}}{12}\right)$
Month (cos)	`cos_month`	$\cos\left(\frac{2\pi\cdot\text{month}}{12}\right)$

Note: Sentinel-2 (S-2) band wavelengths are: B2 = Blue (492 nm), B3 = Green (560 nm), B4 = Red (665 nm), B5 = Red-edge 1 (RE1, 704 nm), B6 = Red-edge 2 (RE2, 740 nm), B7 = Red-edge 3 (RE3, 783 nm), B8 = Near-infrared (NIR, 833 nm), B8A = Narrow-NIR (865 nm), B11 = SWIR1 (1614 nm), B12 = SWIR2 (2202 nm).

Additionally, a seasonal indicator was encoded using the survey month as sine and cosine components. This captures, seasonal variation without an artificial discontinuity between December and January.

References

[1] T. T. Nguyen, T. D. Pham, C. T. Nguyen, J. Delfos, R. Archibald, K. B. Dang, N. B. Hoang, W. Guo, and H. H. Ngo. A novel intelligence approach based active and ensemble learning for agricultural soil organic carbon prediction using multispectral and sar data fusion. *Science of the Total Environment*, 804:150187, 2022. https://doi.org/10.1016/j.scitotenv.2021.150187 doi:10.1016/j.scitotenv.2021.150187.